

Nama : Frederico Steven Kwok

NPM : 242310037

Kelas : TI-24-PA1

Mata Kuliah : Lab. Pemrograman Perangkat Bergerak

Tugas : Pertemuan 7 – Tugas dan Latihan

Github : <https://github.com/RicoSteven120206/PPB-Praktikum-242310037.git>

Pada tugas praktikum ini, akan mengimplementasikan pemanfaatan fitur perangkat (device features) dalam aplikasi mobile yang telah dikembangkan sebelumnya. Fokus utama pada bab ini adalah bagaimana aplikasi dapat berinteraksi langsung dengan kemampuan perangkat seperti kamera, media library, dan sistem pemindaian QR Code.

Mahasiswa diminta untuk mengembangkan dua fitur utama, yaitu:

Latihan 1:

Membuat Halaman Profile Yang Dapat Memperbaharui Data Prbadi dan Avatar Pengguna

⇒ Membuat tampilan untuk halaman profile yang dapat memperbarui data dan avatar, berikut Source Code untuk pengembangannya.

Source Code pada main-apps/(tabs)/profile.jsx

```
import { Ionicons } from "@expo/vector-icons";
import * as ImagePicker from "expo-image-picker";
import { useState } from "react";
import {
  Alert,
  Image,
  ScrollView,
  Text,
  TextInput,
  TouchableOpacity,
  View,
} from "react-native";
import { SafeAreaView } from "react-native-safe-area-context";
import { color_list, styles } from "../../styles/StyleApps";
```

```
export default function ProfileScreen() {
  const [avatar, setAvatar] = useState(null);
  const [isEditing, setIsEditing] = useState(false);

  const [profile, setProfile] = useState({
    name: "Frederico Steven Kwok",
    email: "frederico.steven@example.com",
    bio: "Pecinta buku fiksi dan non-fiksi. Selalu mencari cerita baru.",
  });

  const [tempProfile, setTempProfile] = useState(profile);

  const pickImage = async () => {
    const { status } = await
    ImagePicker.requestMediaLibraryPermissionsAsync();

    if (status !== "granted") {
      Alert.alert(
        "Izin ditolak",
        "Aplikasi membutuhkan akses galeri untuk mengubah foto profil."
      );
      return;
    }

    const result = await ImagePicker.launchImageLibraryAsync({
      mediaTypes: ImagePicker.MediaTypeOptions.Images,
      allowsEditing: true,
      aspect: [1, 1],
      quality: 0.7,
    });

    if (!result.canceled) {
      setAvatar(result.assets[0].uri);
    }
  }
}
```

```

    }
  };

const handleSave = () => {
  if (!tempProfile.name.trim()) {
    Alert.alert("Nama tidak boleh kosong", "Silakan isi nama Anda.");
    return;
  }
  setProfile(tempProfile);
  setIsEditing(false);
  Alert.alert("Berhasil", "Profil Anda telah diperbarui.");
};

const handleCancel = () => {
  setTempProfile(profile);
  setIsEditing(false);
};

return (
  <SafeAreaView style={styles.container}>
    <ScrollView showsVerticalScrollIndicator={false}>
      <View style={{ alignItems: "center", marginTop: 20 }}>
        <TouchableOpacity onPress={isEditing ? pickImage : undefined}>
          <View style={{ position: "relative" }}>
            {avatar ? (
              <Image
                source={{ uri: avatar }}
                style={{ width: 110, height: 110, borderRadius: 55 }}
              />
            ) : (
              <View
                style={{
                  width: 110,

```

```

        height: 110,
        borderRadius: 55,
        backgroundColor: color_list.green_light,
        justifyContent: "center",
        alignItems: "center",
    }}
>
<Icons
  name="person"
  size={50}
  color={color_list.green}
/>
</View>
)}
{isEditing && (
  <View
    style={{
      position: "absolute",
      bottom: 0,
      right: 0,
      width: 32,
      height: 32,
      borderRadius: 16,
      backgroundColor: color_list.green,
      justifyContent: "center",
      alignItems: "center",
      borderWidth: 2,
      borderColor: "#fff",
    }}
  >
    <Icons name="camera" size={16} color="#fff" />
  </View>
)}

```

```

</View>
</TouchableOpacity>

{!isEditing && (
  <◇>
    <Text style={{ fontSize: 20, fontWeight: "bold", marginTop: 12 }}>
      {profile.name}
    </Text>
    <Text style={{ color: "gray", marginTop: 4 }}>{profile.email}</Text>
  </>
)}
</View>

<View style={{ marginTop: 30 }}>
  <FieldRow
    label="Nama Lengkap"
    value={isEditing ? tempProfile.name : profile.name}
    editable={isEditing}
    onChangeText={(text) =>
      setTempProfile((prev) => ({ ...prev, name: text })))
    }
  />
  <FieldRow
    label="Email"
    value={isEditing ? tempProfile.email : profile.email}
    editable={isEditing}
    keyboardType="email-address"
    onChangeText={(text) =>
      setTempProfile((prev) => ({ ...prev, email: text })))
    }
  />
  <FieldRow
    label="Bio"

```

```

        value={isEditing ? tempProfile.bio : profile.bio}
        editable={isEditing}
        multiline
        onChangeText={(text) =>
            setTempProfile((prev) => ({ ...prev, bio: text })))
        }
    />
</View>

<View style={{ marginTop: 30, gap: 10 }}>
    {isEditing ? (
        <
            <TouchableOpacity
                style={{
                    backgroundColor: color_list.green,
                    paddingVertical: 14,
                    borderRadius: 12,
                    alignItems: "center",
                }}
                onPress={handleSave}
            >
                <Text style={{ color: "#fff", fontWeight: "bold", fontSize: 16 }}>
                    Simpan Perubahan
                </Text>
            </TouchableOpacity>
            <TouchableOpacity
                style={{
                    paddingVertical: 14,
                    borderRadius: 12,
                    alignItems: "center",
                    borderWidth: 1,
                    borderColor: "gray",
                }}
            >

```

```

        onPress={handleCancel}
      >
        <Text style={{ color: "gray", fontWeight: "600" }}>Batal</Text>
      </TouchableOpacity>
    </>
  ): (
    <TouchableOpacity
      style={{
        backgroundColor: color_list.green,
        paddingVertical: 14,
        borderRadius: 12,
        alignItems: "center",
        flexDirection: "row",
        justifyContent: "center",
        gap: 8,
      }}
      onPress={() => setIsEditing(true)}
    >
      <Icons name="create-outline" size={20} color="#fff" />
      <Text style={{ color: "#fff", fontWeight: "bold", fontSize: 16 }}>
        Edit Profil
      </Text>
    </TouchableOpacity>
  )}
</View>
</ScrollView>
</SafeAreaView>
);
}

const FieldRow = ({ label, value, editable, onChangeText, multiline,
keyboardType }) => (
  <View style={{ marginBottom: 18 }}>

```

```

        <Text style={{ fontSize: 13, color: "gray", marginBottom: 6
    }}>{label}</Text>
    {editable ? (
      <TextInput
        value={value}
        onChangeText={onChangeText}
        multiline={multiline}
        keyboardType={keyboardType}
        style={{
          borderWidth: 1,
          borderColor: "#ddd",
          borderRadius: 10,
          padding: 12,
          fontSize: 15,
          minHeight: multiline ? 80 : undefined,
          textAlignVertical: multiline ? "top" : "center",
        }}
      />
    ) : (
      <Text style={{ fontSize: 15, color: "#1e293b" }}>{value}</Text>
    )}
  </View>
);

```

Latihan 2:

Membuat fitur QR Scanner Untuk Membaca QR Code Sebuah Buku dan Menampilkan Detailnya

⇒ Berikut Source Code untuk Fitur QR Scanner untuk Membaca sebuah Buku hanya Scan Buku untuk mendapatkan Detail Buku.

Source Code pada main-apps/(tabs)/qr-scanner.jsx

```

import { Icons } from "@expo/vector-icons";
import { CameraView, useCameraPermissions } from "expo-camera";
import { useRouter } from "expo-router";
import { useState } from "react";

```

```
import { Alert, StyleSheet, Text, TouchableOpacity, View } from "react-native";
import { SafeAreaView } from "react-native-safe-area-context";
import ListBook from "../constants/list_book";
import { color_list } from "../styles/StyleApps";

export default function QRScannerScreen() {
  const router = useRouter();
  const [permission, requestPermission] = useCameraPermissions();
  const [scanned, setScanned] = useState(false);

  const handleBarcodeScanned = ({ data }) => {
    if (scanned) return;
    setScanned(true);

    const book = ListBook.find((b) => b.id.toString() === data.toString());

    if (book) {
      router.replace(`/books/${book.id}`);
    } else {
      Alert.alert(
        "QR Code Tidak Dikenali",
        "Buku dengan kode ini tidak ditemukan dalam katalog.",
        [{ text: "Coba Lagi", onPress: () => setScanned(false) }]
      );
    }
  };

  if (!permission) {
    return <View style={qrStyles.container} />;
  }

  if (!permission.granted) {
    return (
```

```

<SafeAreaView style={[qrStyles.container, qrStyles.centerAll]}>
  <Icons name="camera-outline" size={64} color={color_list.green} />
  <Text style={qrStyles.permissionText}>
    Aplikasi membutuhkan akses kamera untuk memindai QR Code buku
  </Text>
  <TouchableOpacity
    style={qrStyles.permissionButton}
    onPress={requestPermission}
  >
    <Text style={qrStyles.permissionButtonText}>Izinkan Akses
Kamera</Text>
  </TouchableOpacity>
</SafeAreaView>
);
}

return (
  <View style={qrStyles.container}>
    <CameraView
      style={StyleSheet.absoluteFillObject}
      facing="back"
      barcodeScannerSettings={{ barcodeTypes: ["qr"] }}
      onBarcodeScanned={scanned ? undefined : handleBarcodeScanned}
    />

    <SafeAreaView style={qrStyles.overlay}>
      <TouchableOpacity
        style={qrStyles.backButton}
        onPress={() => router.back()}
      >
        <Icons name="arrow-back" size={24} color="#fff" />
      </TouchableOpacity>
    </SafeAreaView>
  </View>
);
}

```

```

    <View style={qrStyles.scanFrameContainer}>
      <View style={qrStyles.scanFrame} />
      <Text style={qrStyles.scanHint}>
        Arahkan kamera ke QR Code pada sampul buku
      </Text>
    </View>
  </SafeAreaView>
</View>
);
}

```

```

const qrStyles = StyleSheet.create({
  container: {
    flex: 1,
    backgroundColor: "black",
  },
  centerAll: {
    justifyContent: "center",
    alignItems: "center",
    paddingHorizontal: 30,
  },
  permissionText: {
    textAlign: "center",
    marginTop: 16,
    marginBottom: 20,
    fontSize: 15,
    color: "#333",
  },
  permissionButton: {
    backgroundColor: color_list.green,
    paddingHorizontal: 24,
    paddingVertical: 12,
    borderRadius: 30,
  },
});

```

```
    },
    permissionButtonText: {
      color: "#fff",
      fontWeight: "bold",
    },
    overlay: {
      flex: 1,
      justifyContent: "space-between",
    },
    backButton: {
      margin: 20,
      width: 44,
      height: 44,
      borderRadius: 22,
      backgroundColor: "rgba(0,0,0,0.5)",
      justifyContent: "center",
      alignItems: "center",
    },
    scanFrameContainer: {
      alignItems: "center",
      marginBottom: 80,
    },
    scanFrame: {
      width: 250,
      height: 250,
      borderWidth: 3,
      borderColor: color_list.orange,
      borderRadius: 20,
      backgroundColor: "transparent",
    },
    scanHint: {
      color: "#fff",
      marginTop: 20,
```

```
fontSize: 14,  
textAlign: "center",  
paddingHorizontal: 30,  
},  
});
```

Latihan 3:

Mengimplementasikan fitur perangkat output audio (speaker) yang tidak perlu meminta izin akses perangkat. Buatlah seperti detail buku seperti dibawah ini:

Pada halaman detail buku ini, dapat diubah menjadi audio-book dengan memanfaatkan teknologi text-to-speech.

Manfaatkanlah library package expo-speech dalam mengimplementasikan hal ini. Dimana mengubah sebuah text menjadi sebuah audio dan dikeluarkan melalui speaker device.

Dihalaman ini audio-book dapat di play, pause ataupun stop. Dan memberikan highlight pada text yang sedang dibacakan atau dikonversikan kedalam audio.

⇒ Berikut Source Code yang dikembangkan pada Detail Buku untuk output dari Audio dan Speaker untuk membaca otomatis, dan user hanya mendengarkan saja ketika di play.

Source Code pada /components/books/[id].jsx

```
import { Ionicons } from "@expo/vector-icons";  
import { useLocalSearchParams, useRouter } from "expo-router";  
import {  
  Image,  
  ScrollView,  
  StatusBar,  
  StyleSheet,  
  Text,  
  TouchableOpacity,  
  View,  
} from "react-native";  
import { SafeAreaView } from "react-native-safe-area-context";  
  
import ListBook from "../../constants/list_book";  
import AudioBookPlayer from "../AudioBookPlayer";
```

```

export default function BookDetailScreen() {
  const { id } = useLocalSearchParams();
  const router = useRouter();

  const book = ListBook.find((b) => b.id.toString() === id);

  if (!book) {
    return (
      <SafeAreaView style={[styles.container, styles.centerAll]}>
        <Text style={styles.errorText}>Buku tidak ditemukan</Text>
        <TouchableOpacity
          onPress={() => router.back()}
          style={{ marginTop: 20 }}
        >
          <Text style={{ color: "white" }}>Kembali</Text>
        </TouchableOpacity>
      </SafeAreaView>
    );
  }

  return (
    <SafeAreaView style={styles.container}>
      <StatusBar barStyle="light-content" backgroundColor="#1a2228" />

      <View style={styles.header}>
        <TouchableOpacity
          onPress={() => router.back()}
          style={styles.circleBtn}
        >
          <Ionicons name="arrow-back" size={24} color="#2D3748" />
        </TouchableOpacity>
        <View style={styles.headerRight}>

```

```

        <TouchableOpacity style={styles.circleBtn}>
            <Icons name="heart" size={24} color="#E2E8F0" />
        </TouchableOpacity>
        <TouchableOpacity style={styles.circleBtn}>
            <Icons name="share-social" size={22} color="#2D3748" />
        </TouchableOpacity>
    </View>
</View>

<ScrollView
    showsVerticalScrollIndicator={false}
    contentContainerStyle={styles.scrollContent}
>
    <Image source={book.img} style={styles.coverImage} />

    <Text style={styles.title}>{book.title.toUpperCase()}</Text>
    <Text style={styles.author}>{book.author}</Text>

    <View style={styles.ratingRow}>
        <Icons name="star" size={16} color="#EAB308" />
        <Text style={styles.ratingText}>{book.rating} / 5.0</Text>
    </View>

    {/* Bagian ini yang diganti — synopsisText polos jadi AudioBookPlayer
    */}

    <View style={styles.synopsisBox}>
        <Text style={styles.synopsisLabel}>SINOPSIS</Text>
        <AudioBookPlayer text={book.sinopsis} />

        {book.story && (
            <
                <Text style={[styles.synopsisLabel, { marginTop: 20 }]}>
                    STORY

```

```

        </Text>
        <AudioBookPlayer text={book.story} />
      </>
    )}
  </View>
</ScrollView>

<View style={styles.footer}>
  {book.is_free ? (
    <TouchableOpacity style={[styles.btnAction, styles.btnFree]}>
      <Ionicons name="book-outline" size={20} color="#1E293B" />
      <Text style={styles.txtFree}>Read Book</Text>
    </TouchableOpacity>
  ) : (
    <TouchableOpacity style={[styles.btnAction, styles.btnPremium]}>
      <Ionicons name="card-outline" size={20} color="#FFF" />
      <Text style={styles.txtPremium}>Subscribe</Text>
    </TouchableOpacity>
  )}
</View>
</SafeAreaView>
);
}

const styles = StyleSheet.create({
  container: {
    flex: 1,
    backgroundColor: "#1a2228",
  },
  centerAll: {
    justifyContent: "center",
    alignItems: "center",
  },
});

```

```
errorText: {
  color: "#FFF",
  fontSize: 18,
  fontWeight: "bold",
},
header: {
  flexDirection: "row",
  justifyContent: "space-between",
  paddingHorizontal: 20,
  paddingTop: 10,
  paddingBottom: 20,
},
headerRight: {
  flexDirection: "row",
  gap: 15,
},
circleBtn: {
  width: 44,
  height: 44,
  borderRadius: 22,
  backgroundColor: "#FFF",
  justifyContent: "center",
  alignItems: "center",
},
scrollContent: {
  alignItems: "center",
  paddingHorizontal: 20,
  paddingBottom: 40,
},
coverImage: {
  width: 200,
  height: 200,
  borderRadius: 16,
```

```
marginBottom: 25,
resizeMode: "cover",
},
title: {
  color: "#FFF",
  fontSize: 22,
  fontWeight: "bold",
  textAlign: "center",
  marginBottom: 8,
},
author: {
  color: "#E2E8F0",
  fontSize: 16,
  fontWeight: "500",
  marginBottom: 10,
},
ratingRow: {
  flexDirection: "row",
  alignItems: "center",
  marginBottom: 30,
},
ratingText: {
  color: "#EAB308",
  fontSize: 14,
  fontWeight: "bold",
  marginLeft: 6,
},
synopsisBox: {
  width: "100%",
},
synopsisLabel: {
  color: "#FFF",
  fontSize: 18,
```

```
fontWeight: "bold",
marginBottom: 10,
},
synopsisText: {
  color: "#CBD5E1",
  fontSize: 15,
  lineHeight: 24,
  textAlign: "justify",
},
footer: {
  padding: 20,
  paddingBottom: 30,
  backgroundColor: "transparent",
},
btnAction: {
  flexDirection: "row",
  width: "100%",
  paddingVertical: 16,
  borderRadius: 30,
  justifyContent: "center",
  alignItems: "center",
  gap: 10,
},
btnPremium: {
  backgroundColor: "#EAB308",
},
txtPremium: {
  color: "#FFF",
  fontSize: 18,
  fontWeight: "bold",
},
btnFree: {
  backgroundColor: "#FFF",
```

```

    },
    txtFree: {
      color: "#1E293B",
      fontSize: 18,
      fontWeight: "bold",
    },
  });

```

Source Code pada /components/books/AudioBookPlayer.jsx

```

import { Ionicons } from "@expo/vector-icons";
import * as Speech from "expo-speech";
import { useEffect, useState } from "react";
import { Text, TouchableOpacity, View } from "react-native";
import { color_list } from "../styles/StyleApps";

const AudioBookPlayer = ({ text }) => {
  const [status, setStatus] = useState("idle");
  const [boundary, setBoundary] = useState({ charIndex: 0, charLength: 0 });

  useEffect(() => {
    return () => {
      Speech.stop();
    };
  }, []);

  const handlePlay = () => {
    if (status === "paused") {
      Speech.resume();
      setStatus("playing");
      return;
    }

    setStatus("playing");
    Speech.speak(text, {

```

```
language: "id-ID",
pitch: 1.0,
rate: 0.95,
onBoundary: (event) => {
  setBoundary({
    charIndex: event.charIndex,
    charLength: event.charLength,
  });
},
onDone: () => {
  setStatus("idle");
  setBoundary({ charIndex: 0, charLength: 0 });
},
onStopped: () => {
  setStatus("idle");
  setBoundary({ charIndex: 0, charLength: 0 });
},
onError: () => {
  setStatus("idle");
},
});
};
```

```
const handlePause = () => {
  Speech.pause();
  setStatus("paused");
};
```

```
const handleStop = () => {
  Speech.stop();
  setStatus("idle");
  setBoundary({ charIndex: 0, charLength: 0 });
};
```

```

const renderHighlightedText = () => {
  if (status === "idle" || boundary.charLength === 0) {
    return <Text style={{ color: "#CBD5E1", fontSize: 15, lineHeight: 24
}}>{text}</Text>;
  }

  const before = text.slice(0, boundary.charIndex);
  const current = text.slice(
    boundary.charIndex,
    boundary.charIndex + boundary.charLength
  );
  const after = text.slice(boundary.charIndex + boundary.charLength);

  return (
    <Text style={{ fontSize: 15, lineHeight: 24 }}>
      <Text style={{ color: "#CBD5E1" }}>{before}</Text>
      <Text
        style={{
          color: "#1a2228",
          backgroundColor: color_list.orange,
          fontWeight: "bold",
        }}
      >
        {current}
      </Text>
      <Text style={{ color: "#CBD5E1" }}>{after}</Text>
    </Text>
  );
};

return (
  <View style={{ marginTop: 20 }}>

```

```

<View
  style={{
    flexDirection: "row",
    alignItems: "center",
    backgroundColor: "#232d36",
    borderRadius: 16,
    padding: 16,
    marginBottom: 16,
  }}
>
  {status === "playing" ? (
    <TouchableOpacity
      onPress={handlePause}
      style={{
        width: 44,
        height: 44,
        borderRadius: 22,
        backgroundColor: color_list.orange,
        justifyContent: "center",
        alignItems: "center",
      }}
    >
      <Ionicons name="pause" size={22} color="#fff" />
    </TouchableOpacity>
  ) : (
    <TouchableOpacity
      onPress={handlePlay}
      style={{
        width: 44,
        height: 44,
        borderRadius: 22,
        backgroundColor: color_list.orange,
        justifyContent: "center",

```

```

        alignItems: "center",
      }}
    >
    <Icons name="play" size={22} color="#fff" />
  </TouchableOpacity>
)}

<TouchableOpacity
  onPress={handleStop}
  style={{
    width: 44,
    height: 44,
    borderRadius: 22,
    backgroundColor: "#3a4650",
    justifyContent: "center",
    alignItems: "center",
    marginLeft: 12,
  }}
>
  <Icons name="stop" size={20} color="#fff" />
</TouchableOpacity>

<Text style={{ color: "#CBD5E1", marginLeft: 14, fontSize: 13 }}>
  {status === "playing"
    ? "Sedang membaca..."
    : status === "paused"
    ? "Dijeda"
    : "Audio-Book"}
</Text>
</View>

{renderHighlightedText()}
</View>

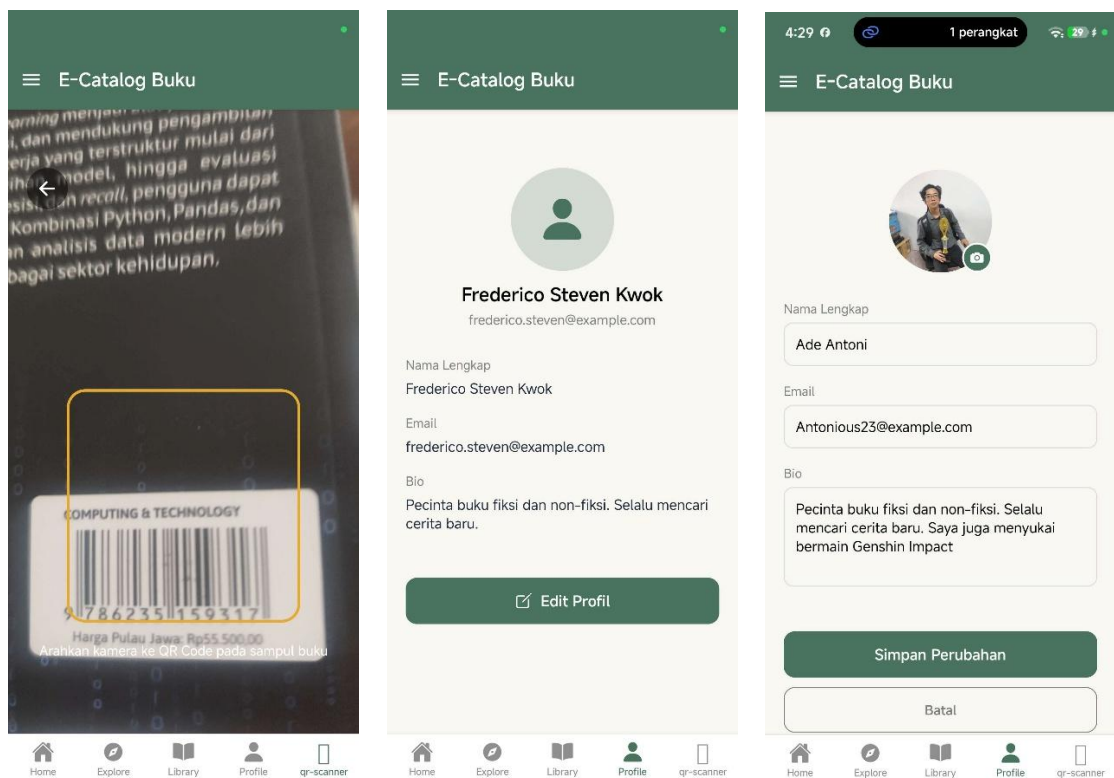
```

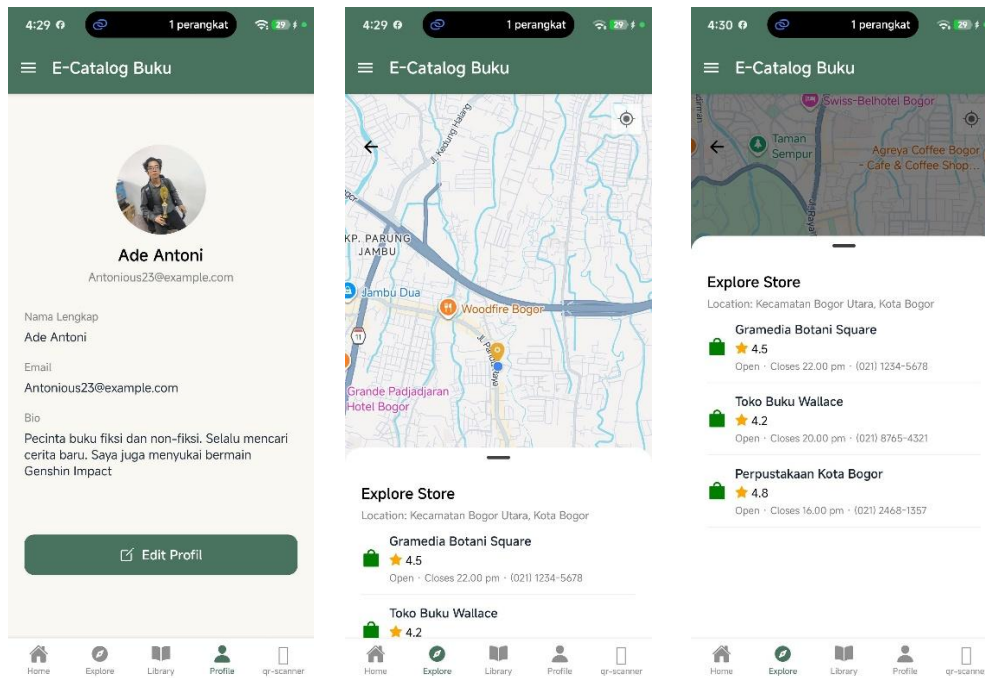
```
);  
};  
  
export default AudioBookPlayer;
```

Latihan 4: Laporan Singkat

1. Buatlah dokumentasi dalam mengerjakan soal Latihan 1 hingga Latihan 3, dengan memberikan Screenshot baik dari sisi aplikasi ataupun code yang sudah dimodifikasi dengan penjelasan dan alur yang baik dan benar. Dokumentasi disimpan dalam bentuk PDF dan ditaruh dalam project react expo yang telah anda kerjakan.

⇒ Dokumentasi pada Output Dari Pengerjaan





2. Upload project tersebut pada GITHUB yang telah dibuat sebelumnya di repository "PPB-Praktikum-NPM", namun commit dengan nama "Tugas Praktikum-7".
3. Pengumpulan Tugas Latihan Praktikum dikumpulkan selama 1 minggu kedepan sejak hari ini (sesuai dengan jadwal praktikum).

LINK REPOSITORY

<https://github.com/RicoSteven120206/PPB-Praktikum-242310037.git>